Fiscal Year:	FY 2022	**Task Last Updated:**	FY 06/27/2022
PI Name:	Stankovic, Aleksandra Ph.D.		
Project Title:	Modeling Individual and Team Problem-Solving Skills for the Establishment of Training Metrics, Standards, and Guidelines for Long-Duration Space Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	astankovic1@mgh.harvard.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	908-391-1177
Organization Name:	Massachusetts General Hospital/Harvard Medical School		
PI Address 1:	Human Performance Laboratory		
PI Address 2:	149 13th St		
PI Web Page:			
City:	Charlestown	**State:**	MA
Zip Code:	02129-2020	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC018N0001-HHCHFBP: Human Health Countermeasures, Human Factors, Behavioral Performance. Appendix D
Start Date:	03/09/2020	**End Date:**	03/08/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Robinson, Stephen Ph.D. (University of California, Davis) Strangman, Gary (Massachusetts General Hospital/Harvard Medical School) Caldwell, Barrett (Purdue University)		
Grant/Contract No.:	80NSSC20K0844		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Motivation: Future human lunar exploration will necessitate that astronaut crews work autonomously to solve unexpected and time-critical problems, and will demand increased cooperation between humans and robotic/autonomous systems. While the ability of these multi-agent teams to adapt to unforeseen and off-nominal situations will be integral to mission success and safety, frameworks and recommendations for cultivating team-based problem-solving are currently lacking. Goals: This project aims to characterize team problem-solving strategies for future long duration space exploration tasks, specifically in the context of lunar and cislunar operations. This work will generate findings that will help shape team-based concepts of operations and functional requirements, will contribute towards the development of a set of validated measures and metrics for assessing team problem-solving, and will produce an approach to modeling dynamic team problem-solving performance over time, with the goal of helping determine training effectiveness. Outcomes of this study will also help shape future decision and execution support systems. This project has 3 integrated specific aims: Aim 1: Through literature review and in collaboration with teaming subject matter experts, we will (a) identify key teaming and problem-solving knowledge gaps and attributes of interest, and will (b) develop an approach for the in-situ assessment and characterization of individual and team-based problem solving using operationally-deployable metrics and measures. (Completed.) Aim 2: Through data collection in Earth-bound analogs for lunar/cislunar activities, we will examine attributes contributing to individual and team problem-solving effectiveness, and will characterize how such skills are developed and used across a variety of operational tasks, under both normal and emergency conditions. (Currently in progress.) Aim 3: Leveraging the insights gained in Aims 1 & 2, we will develop a set of team-based concepts of operations and functional requirements, which may ultimately be used to inform future training guidelines and recommendations for individual and team problem-solving, including for the effective interface with on-board support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are undertaking an operationally-relevant assessment of team problem-solving skills in the context of upcoming cislunar exploration, with the ultimate goal of contributing towards the establishment of future autonomous team-based concepts of operations and functional design requirements for system development, operations, and training.
Task Progress:	Technical Approach: Using relevant Earth-bound analogs for anticipated lunar/cislunar activities across 4 key domains (construction/repair, science return, medical operations, and emergency response), we will, through contextual inquiry, (1) determine how problem-solving skills are developed and used across a variety of operational tasks and situations relevant to future long-duration spaceflight exploration, and (2) identify which team and human-autonomous systems collaboration skills are most important to performing tasks in both normal and emergency conditions. Research Phases: The project includes three research phases: (1) literature review and translational methodology development (concluded); (2) in-situ data collection (e.g., contextual inquiry, observation, interviews, surveys) and data analysis (to commence in Fall 2022); and (3) development of models, concepts of operations, and function requirements for effective teaming in lunar and cislunar operations. Methods: The first phase of this investigation involved an extensive literature review that resulted in the identification of processes and outcomes of interest, along with key knowledge gaps and needs (manuscript currently under review). This initial definition phase has guided our methodology development, and the refinement of our data collection tools (e.g., surveys, interview guides), which we will deploy in the next phase of our investigation. In this second phase, we will examine team-based problem-solving across several analog environments through a combination of semi-structured interviews, surveys, embedded task observation, and contextual inquiry. These methods are ideal for this type of investigation because they offer higher fidelity examinations of operational conditions than laboratory testing, and can be highly customizable to different situations and contexts. Through this investigation, we will develop organizational and hierarchical breakdowns to characterize the relationships among team members, and will examine how factors such as tasks, roles, operational situations, work phases, experience, training, collaboration with autonomous tools, and other key variables identified through the literature analysis phase relate to the formation and appraisal of various team-based skills and needs. We plan to conduct field research at these sites over a period of 9-12 months, with roughly 1-3 months engagement per site across 5 sites, each corresponding to one of the 4 domain types identified (construction/repair, science return, medical operations, and emergency response), plus one multi-role extreme environment crew setting (e.g., Antarctic crew). During the field research, data collection will involve (a) weekly surveys (incorporating modified questionnaires developed as part of the NASA Human Factors and Behavioral Performance (HFBP) Exploration Measures for Team Risks) focusing on team performance, cohesion, team processes, group living, and social support), (b) 3 semi-structured interviews (conducted at the start, mid, and endpoints of the observation period) to discuss in detail problem-solving processes with crewmembers, along with outcomes and related attributes in the context of their work, (c) and embedded task observation. Following on from our characterization of contextually-specific team skills across these field sites, we will assess (qualitatively and quantitatively) the data collected to determine the relationship between individual and team performance attributes and problem-solving skills, and will characterize the change in skill demand over time (e.g., the project lifecycle). Through this translational work, we will explore what insights can be gained from examining such Earth-bound operational analog sites, and will leverage them for developing recommendations that can be applied to the astronaut population. By understanding the various elements and dimensions of team problem-solving that are relevant to anticipated lunar and cislunar activities, we will translate the findings of our investigation in Earth-bound analogs to operationally significant conclusions that can be applied to the tracking and assessment of team-based performance in long-duration spaceflight. For this, we will leverage systems engineering methodologies to model effective team performance outcomes and generate a framework for promoting team functioning resilience through pre-mission training and optimal interface with on-board support systems. These will provide the basis for a series of matrixed task-specific recommendations that will be developed with a consideration for integration into pre-mission training flows, as well as dynamic just-in-time and on-board refresher training programs. Progress Updates: Project accomplishments to date include the following items (representing completion of

	definition-phase exit milestones): literature review, identification and recruitment of teaming expert co-Investigator, identification of subject matter experts for informational interviewing and data collection, updating of project timeline and definition of next phase milestones, completion of the Life Sciences Data Archive (LSDA) data sharing agreement, research protocol definition and drafting of Institutional Review Board(IRB) materials, definition of field site requirements, and early-stage logistical planning for field research. In the next 3 months, we anticipate IRB approval, analog research site finalization, and preparations to embark on field-based data collection.
Bibliography Type:	Description: (Last Updated: 03/18/2026)
Abstracts for Journals and Proceedings	Thoolen S, Harris K, White B, Caldwell B, Robinson S, Strangman G, Stankovic A. "Team problem solving skills for lunar exploration." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022